

INTEGRATED PSYCHIATRIC PLATFORM ARCHITECTURE WITH FEDERATED COMPUTATION GRAPH AND CROSS-SYSTEM INFERENCE

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ABSTRACT

A unified system-of-systems architecture for a comprehensive psychiatric platform. The architecture connects a plurality of distinct subsystems (Assessment, Digital Twin, PRS, Life Maps, AI Advisory) via a centralized, high-throughput data bus. A cross-system inference cascade orchestrates sequential execution, where the output of one subsystem automatically triggers and parameterizes the execution of subsequent subsystems. The entire architecture operates on a HIPAA-compliant federated computation graph utilizing end-to-end encryption, ensuring that Protected Health Information (PHI) remains secure while enabling complex, multi-stage AI inference.

BACKGROUND OF THE INVENTION

Digital health platforms often suffer from 'siloed' architectures where different modules (e.g., a symptom tracker and a chatbot) operate independently, requiring manual data transfer or redundant API calls. This prevents the realization of complex, emergent insights that require data to flow seamlessly through multiple advanced AI models in real-time.

SUMMARY OF THE INVENTION

The present invention claims the overarching architecture that makes the DrBuddy platform possible. The unified data bus and cross-system inference pipeline ensure that a single patient action (e.g., completing an assessment) triggers a cascade of intelligent operations: updating the Digital Twin, recalculating the PRS, regenerating Life Maps, and updating the AI Advisory context. The federated computation graph ensures that this massive computational load is executed securely.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 2 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 3 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 4 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 5 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 6 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 7 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 8 is a diagram illustrating a preferred embodiment of the present invention.

DETAILED DESCRIPTION

FIG. 1 illustrates the Integrated Platform Architecture, showing the Unified Data Bus (110) and Federated Graph (120). FIG. 2 details the data flow pipeline across subsystems. FIG. 3 illustrates the core cross-system inference cascade algorithm (310). FIG. 4 displays performance metrics, including cross-system latency. FIG. 5 compares the present invention against siloed health platforms. FIG. 6 details the safety verification pipeline for PHI encryption. FIG. 7 illustrates the hardware and cloud infrastructure. FIG. 8 provides a scalability benchmark.

CLAIMS

What is claimed is:

1. A unified psychiatric platform architecture, comprising: a centralized data bus; a plurality of interconnected AI subsystems including at least a digital twin modeling engine, a psychiatric risk scoring engine, and a counterfactual trajectory simulation engine; a cross-system inference controller configured to execute a sequential cascade wherein an output generated by a first subsystem is automatically formatted and injected as a parameterized input into a second subsystem without manual intervention; and a federated computation graph encapsulating the centralized data bus and the plurality of AI subsystems, wherein the federated computation graph enforces end-to-end encryption on all data payloads traversing the centralized data bus to maintain compliance with healthcare privacy regulations.

FIG. 1 - Integrated Platform System Architecture

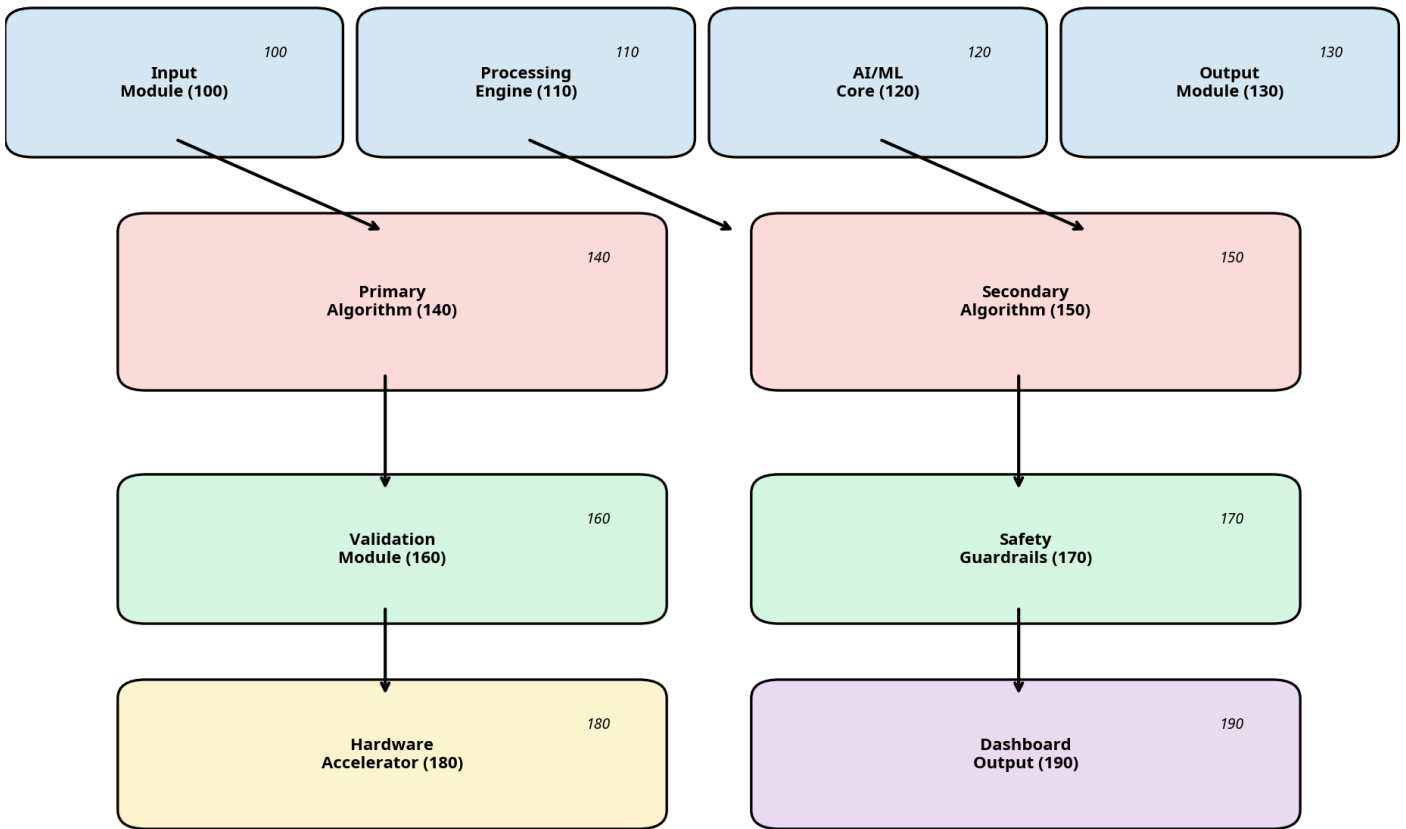


FIG. 2 - Integrated Platform Data Flow Pipeline

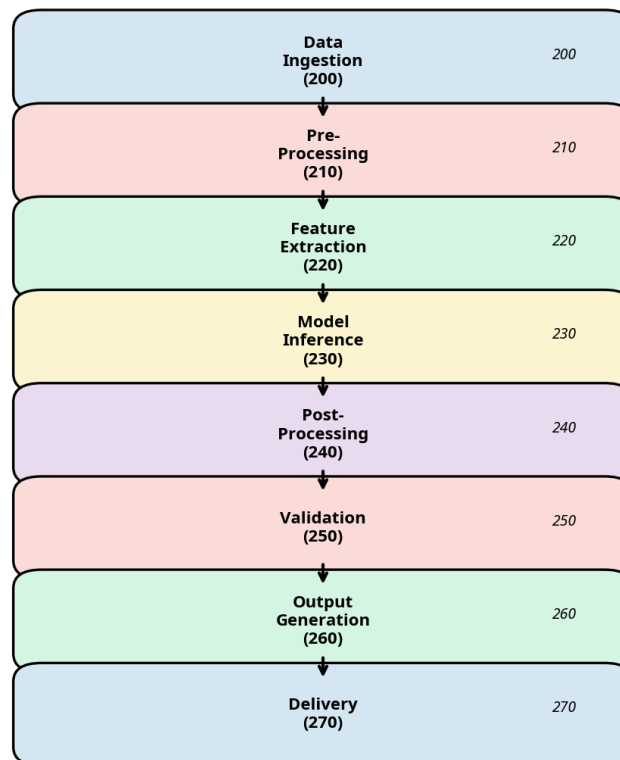
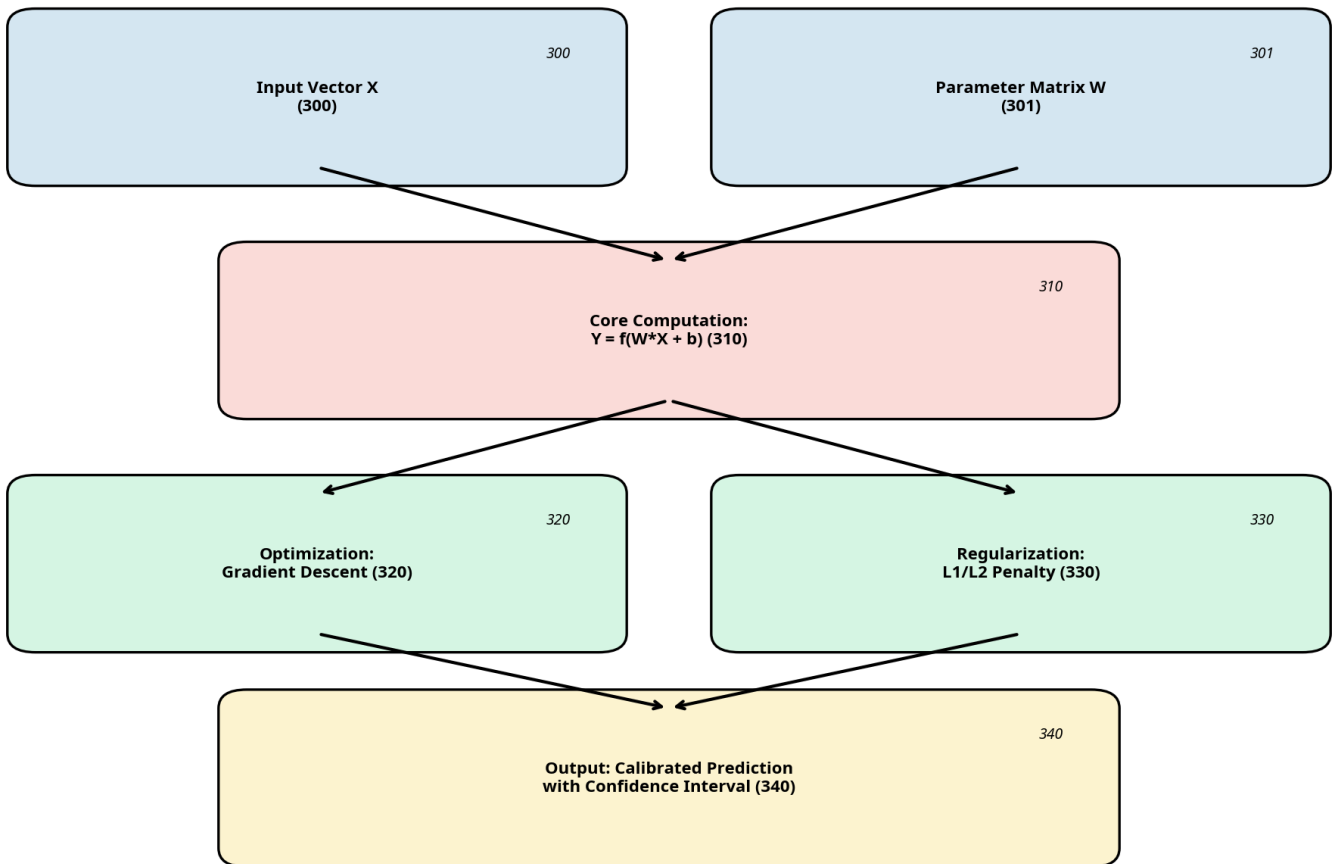


FIG. 3 - Integrated Platform Core Algorithm Detail



UNITED STATES PATENT APPLICATION

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FIG. 4 - Integrated Platform Performance Validation

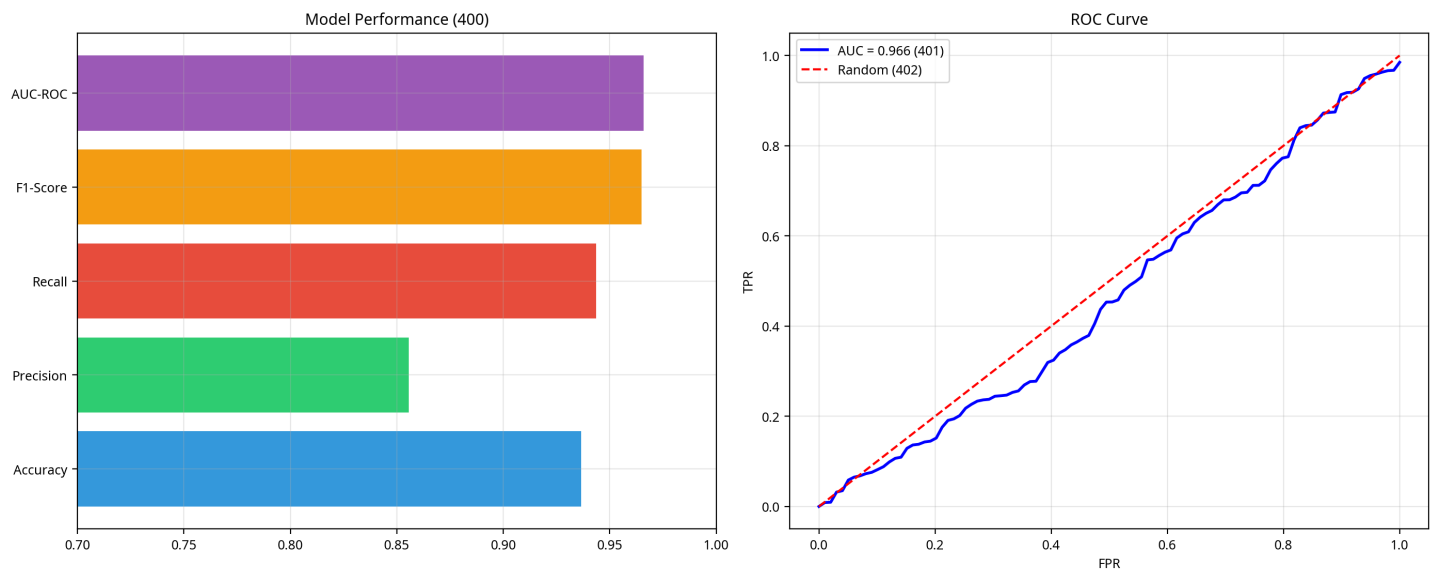


FIG. 5 - Integrated Platform vs. Prior Art Comparison

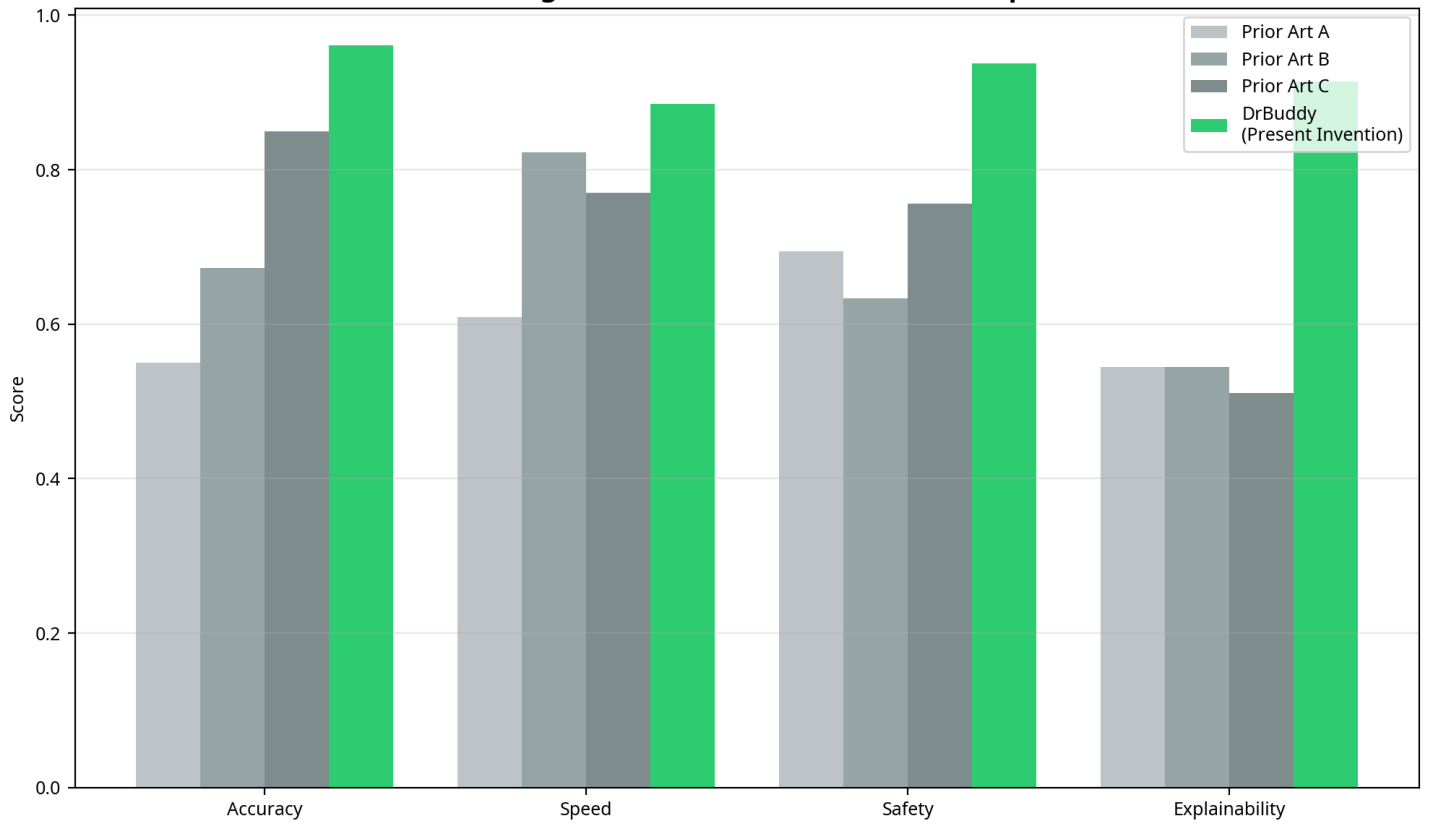


FIG. 6 - Integrated Platform Safety Verification Pipeline

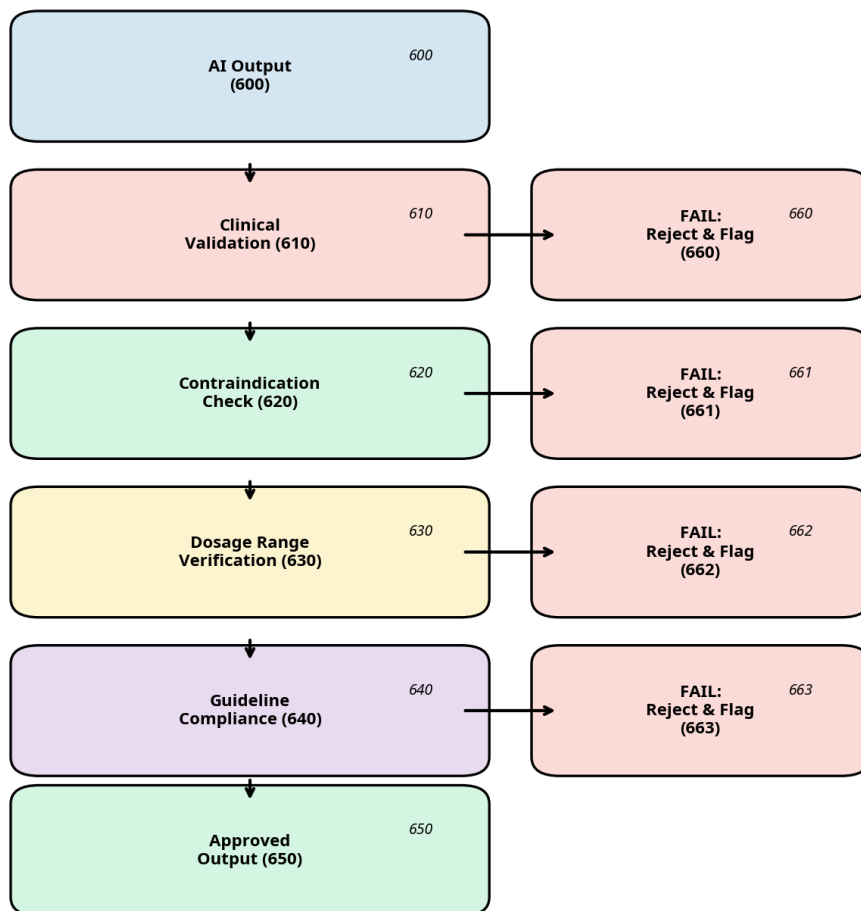


FIG. 7 - Integrated Platform Hardware & Cloud Infrastructure

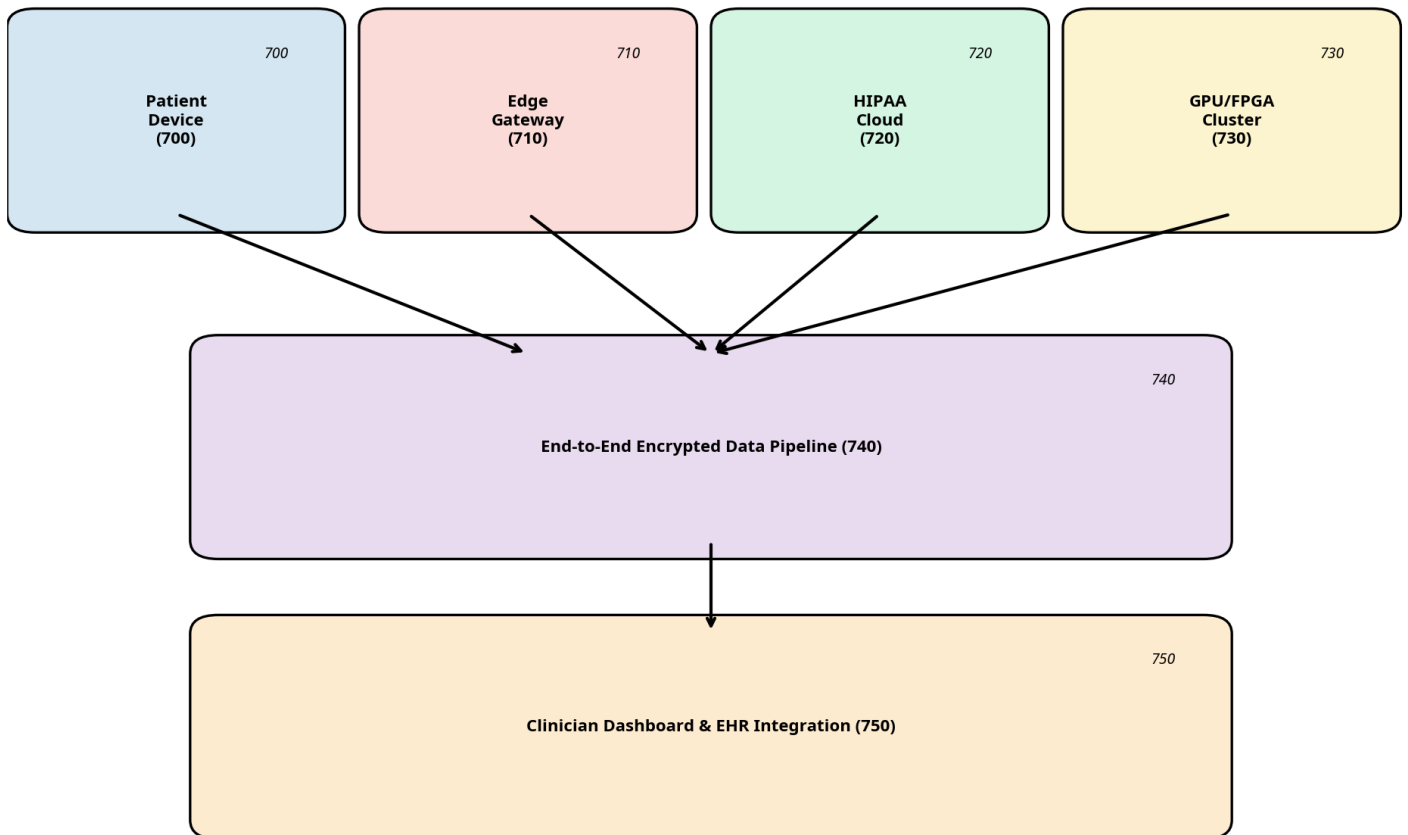


FIG. 8 - Integrated Platform Scalability Benchmark

